

RAPPORT_EXHAUSTIF_ENGINES_Ω — DOCUMENTATION COMPLÈTE

Émis pour COMMANDANT STEEVE-MAX · 2026-04-28T21:25:55.005625+00:00 · 209 engines · BIONIC OS · Articles 1-5

VERDICT INSTITUTIONNEL

Documentation exhaustive de 209 engines sur 5 piliers institutionnels. V30 LOCKED INVIOLE · Registry V30 sealed (41 engines, SHA `27516c9633853974fbb5754f4698a227...`). 122 engines actifs Ω recommandés à la conservation. Recommandations détaillées pour PHASE 4 (à conserver / purger / réactiver / valider).

§1 — STATISTIQUES GLOBALES (Article 1)

Total engines	Actif Ω	Actif legacy
209	122	20
Inactif legacy (à purger)	Inactif V8_INST (à valider)	Inactif orphelin
39	16	12

Distribution par pilier institutionnel

GOUVERNANCE	LEGACY	BIO-SYSTEME	ENVIRONNEMENT	COMPORTEMENT-HUMAIN
95	65	24	20	4
SYSTEME-SENSORIEL				
1				

§2 — V30 LOCKED · CRYPTOGRAPHIE INTÉGRITÉ

Fichier	SHA-256	Taille	Statut
registry_lock_omega.py	fb765b94cc1fd4216c4afa4c0fb72bc1fd8e18fc26b6955db8157b42a26ecb0c	8703 octets	INVIOLE
engine_ia_corridors_omega.py	bcb1e3a6a92304a171978ee7b6be2151e7035c84d8ffc1690839d993be9e39d3	17448 octets	INVIOLE
Registry V30 sealed	27516c9633853974fbb5754f4698a227bf39346e94f274889d4b4ee0398f7e4c	41 engines	SCELLÉ 2026-04-21T05:30:00Z

§3 — RECOMMANDATIONS STRUCTURÉES (Article 3)

A. À CONSERVER (122 modules — actif Ω)

- engines.advanced_geospatial_omega.router
- engines.bio_scoring_omega.router
- engines.bio_scoring_omega.v30_mirror_read_only
- engines.eco_zones_omega.router
- engines.ecoforestry_omega.router
- engines.hydro_topo_omega.router

- engines.legal_time_omega.router
- engines.post_smoothing.anti_regression_omega
- engines.post_smoothing.baseline_registry_omega
- engines.post_smoothing.interzone_omega
- engines.post_smoothing.organic_corridor_smoother
- engines.post_smoothing.p1_preparation
- engines.post_smoothing.predictive_integration
- engines.post_smoothing.renduomega
- engines.post_smoothing.terrain_signals_builder
- engines.post_smoothing.veineux_omega
- engines.predictive_omega.router
- engines.reseau_veineux_omega.external_inflow
- engines.reseau_veineux_omega.router
- engines.terrain_3d_omega.router
- engines.v8_institutional.corridors_vitaux_omega
- engines.v8_institutional.ecological_orchestrator_omega
- engines.v8_institutional.engine_affuts
- engines.v8_institutional.engine_audio_acoustique
- engines.v8_institutional.engine_bio_signes
- engines.v8_institutional.engine_calibration_dynamique_omega
- engines.v8_institutional.engine_calibration_omega
- engines.v8_institutional.engine_canada_omega
- engines.v8_institutional.engine_canopee_thermique_omega
- engines.v8_institutional.engine_carence_nutritionnelle_omega
- engines.v8_institutional.engine_champs_nourriciers_omega
- engines.v8_institutional.engine_climat_futur_omega
- engines.v8_institutional.engine_comportement
- engines.v8_institutional.engine_comportement_avance
- engines.v8_institutional.engine_comportement_biologique_omega
- engines.v8_institutional.engine_connectivite
- engines.v8_institutional.engine_connectivite_ecologique_omega
- engines.v8_institutional.engine_contamination_v2_omega
- engines.v8_institutional.engine_espece_omega
- engines.v8_institutional.engine_forage_qualite_omega
- engines.v8_institutional.engine_frequentation
- engines.v8_institutional.engine_gouvernance_omega
- engines.v8_institutional.engine_habitat_supra
- engines.v8_institutional.engine_heatmap
- engines.v8_institutional.engine_hotspots
- engines.v8_institutional.engine_hydrologie_supra
- engines.v8_institutional.engine_ia_corridors_omega
- engines.v8_institutional.engine_ia_corridors_organic_omega
- engines.v8_institutional.engine_ia_vision_ecologique_omega
- engines.v8_institutional.engine_incertitude_omega
- engines.v8_institutional.engine_influence_lunaire_omega
- engines.v8_institutional.engine_intelligence
- engines.v8_institutional.engine_microclimat_advanced_omega
- engines.v8_institutional.engine_nutrition_v12_supra
- engines.v8_institutional.engine_nutritional_attractiveness_omega
- engines.v8_institutional.engine_optimisation_habitat_omega
- engines.v8_institutional.engine_population_dynamics_omega

- engines.v8_institutional.engine_prediction
- engines.v8_institutional.engine_pression
- engines.v8_institutional.engine_pression_atmospherique_omega
- engines.v8_institutional.engine_psychologie
- engines.v8_institutional.engine_qualite_donnees_omega
- engines.v8_institutional.engine_recettes_salines_omega
- engines.v8_institutional.engine_render_omega
- engines.v8_institutional.engine_risque
- engines.v8_institutional.engine_risques_hydro_omega
- engines.v8_institutional.engine_saisonnalite
- engines.v8_institutional.engine_salines
- engines.v8_institutional.engine_salines_v11_supra
- engines.v8_institutional.engine_sante_physio_omega
- engines.v8_institutional.engine_science_omega
- engines.v8_institutional.engine_score_global
- engines.v8_institutional.engine_sensoriel_vent_odeurs_omega
- engines.v8_institutional.engine_social_structure_omega
- engines.v8_institutional.engine_sol_nutriments_omega
- engines.v8_institutional.engine_sol_supra
- engines.v8_institutional.engine_species_profiles_omega
- engines.v8_institutional.engine_stress_anthropique_omega
- engines.v8_institutional.engine_terrain_cost
- engines.v8_institutional.engine_thermique_microclimat_omega

B. À PURGER (39 modules — inactif legacy)

- engines.bdre.__init__
- engines.bdre.audit_logger
- engines.bdre.fallback_chain
- engines.bdre.quality_scorer
- engines.bdre.router
- engines.bdre.source_registry
- engines.bdre.source_selector
- engines.bdre.waterway_classifier
- engines.corridor_unified.__init__
- engines.corridor_unified.router
- engines.hunt_orchestrator.__init__
- engines.hunt_orchestrator.router
- engines.nutrition_intelligence.__init__
- engines.nutrition_intelligence.router
- engines.nutrition_intelligence.x5100_mineral_score
- engines.nutrition_intelligence.x5200_mineral_recommendation
- engines.nutrition_intelligence.x5300_order_engine
- engines.nutrition_intelligence.x5500_energy_protein
- engines.nutrition_intelligence.x5600_site_guide
- engines.nutrition_intelligence.x5700_cost_engine
- engines.nutrition_intelligence.x5800_recipe_engine
- engines.nutrition_intelligence.x5900_evidence_engine
- engines.nutrition_intelligence.x6010_product_quality_analyzer
- engines.nutrition_intelligence.x6011_market_availability_engine
- engines.nutrition_intelligence.x6012_regulatory_compliance_engine

- engines.nutrition_intelligence.x6020_terrain_solutions
- engines.nutrition_intelligence.x6030_product_ecosystem
- engines.nutrition_intelligence.x7000_supplier_product_engine
- engines.post_smoothing.__init__
- engines.relocation.__init__
- engines.relocation.router
- engines.spatial_engine_v7.__init__
- engines.spatial_engine_v7.router
- engines.supra_advanced.__init__
- engines.supra_advanced.router
- engines.supra_engine_v7.__init__
- engines.supra_engine_v7.router
- engines.terrain_nav.__init__
- engines.weather_v3.__init__

C. À RÉACTIVER (16 modules — V8_INSTITUTIONAL inactif)

- engines.v8_institutional._ARCHIVE_NON_ACTIVE.engine_corridors_legacy_pre_L
- engines.v8_institutional.__init__
- engines.v8_institutional.engine_cameras
- engines.v8_institutional.engine_ia_vision_registry_omega
- engines.v8_institutional.engine_nutrition
- engines.v8_institutional.engine_rendu_omega
- engines.v8_institutional.engine_territoire_anti_regression_omega
- engines.v8_institutional.engines_catalog
- engines.v8_institutional.hotspots_organic_v1
- engines.v8_institutional.phase_omega_secure_lockdown
- engines.v8_institutional.piliers_router
- engines.v8_institutional.salines_organic_v1
- engines.v8_institutional.supra_v8
- engines.v8_institutional.v20_mvt_tiles
- engines.v8_institutional.visual_proof_live_playwright
- engines.v8_institutional.zones_organic_v1

D. ORPHELINS (12 modules — non-importés)

- engines.advanced_geospatial_omega.__init__
- engines.bio_scoring_omega.__init__
- engines.eco_zones_omega.__init__
- engines.ecoforestry_omega.__init__
- engines.hydro_topo_omega.__init__
- engines.legal_time_omega.__init__
- engines.predictive_omega.__init__
- engines.reseau_veineux_omega.__init__
- engines.terrain_3d_omega.__init__
- engines.v8_national.__init__
- engines.v8_national.router
- engines.wildlife_behavior_omega.__init__

E. MANQUANTS POUR VALIDATION (11 engines registry sans marker en code)

- ANTI-CONTAMINATION-INSTITUTIONNEL-Ω

- ENGINE-ALERTE-ANOMALIES-Ω
- ENGINE-CONTAMINATION-Ω-V2
- ENGINE-QUALITE-DONNEES-Ω
- EXPORT-INSTITUTIONNEL-V20-Ω
- FEDERAL-DATASETS-Ω
- SCIENCE-GAPS-DATASETS-Ω
- SELF-AUDIT-ALERTS-Ω
- SLA-BASELINE-30J-Ω
- VISUAL-PROOF-LIVE-Ω
- VISUAL-PROOF-Ω

F. CONFLITS POTENTIELS — doublons par stem (2)

- router → engines/legal_time_omega/router.py · engines/v8_national/router.py · engines/bio_scoring_omega/router.py · engines/predictive_omega/router.py · engines/reseau_veineux_omega/router.py · engines/wildlife_behavior_omega/router.py · engines/advanced_geospatial_omega/router.py · engines/eco_zones_omega/router.py · engines/ecoforestry_omega/router.py · engines/hydro_topo_omega/router.py · engines/terrain_3d_omega/router.py · engines/weather_v3/router.py · engines/bdre/router.py · engines/corridor_unified/router.py · engines/hunt_orchestrator/router.py · engines/nutrition_intelligence/router.py · engines/relocation/router.py · engines/spatial_engine_v7/router.py · engines/supra_advanced/router.py · engines/supra_engine_v7/router.py
- __init__ → engines/v8_institutional/__init__.py · engines/legal_time_omega/__init__.py · engines/v8_national/__init__.py · engines/bio_scoring_omega/__init__.py · engines/predictive_omega/__init__.py · engines/reseau_veineux_omega/__init__.py · engines/wildlife_behavior_omega/__init__.py · engines/advanced_geospatial_omega/__init__.py · engines/eco_zones_omega/__init__.py · engines/ecoforestry_omega/__init__.py · engines/hydro_topo_omega/__init__.py · engines/terrain_3d_omega/__init__.py · engines/bdre/__init__.py · engines/corridor_unified/__init__.py · engines/hunt_orchestrator/__init__.py · engines/nutrition_intelligence/__init__.py · engines/post_smoothing/__init__.py · engines/relocation/__init__.py · engines/spatial_engine_v7/__init__.py · engines/supra_advanced/__init__.py · engines/supra_engine_v7/__init__.py · engines/terrain_nav/__init__.py · engines/weather_v3/__init__.py

§4 — INVENTAIRE DÉTAILLÉ DES 209 ENGINES (Articles 1-2)

ENGINE_NAME	PILIER · PHASE	STATUT	PATH	SHA / SIZE	DESCRIPTION FONCTIONNELLE	COUCHES Ω ÉMISES	APIs EXPOSÉES	RISQUES
router	GOUVERNANCE Ω-X	ACTIF_Ω	engines/legal_time_omega/router.py	3212acc450f2667a...8210o	ENGINE_LEGAL_TIME_Ω — Router FastAPI — X199 ACTIVÉ	—	POST / GET /status	—
anti_regression_omega	GOUVERNANCE Ω	ACTIF_Ω	engines/post_smoothing/anti_regression_omega.py	c644223b4c90fd36...13837o	anti_regression_omega.py — PHASE_X200_P6_ANTI_RÉGRESSION_Ω	—	—	—
baseline_registry_omega	GOUVERNANCE Ω	ACTIF_Ω	engines/post_smoothing/baseline_registry_omega.py	6712bc21dd619fef...4451o	baseline_registry_omega.py — PHASE_XII_SUPRA_CORRIDORS_VEINEUX_Ω_ULTIME_ENFORCEMENT_PO	—	—	—
interzone_omega	GOUVERNANCE Ω	ACTIF_Ω	engines/post_smoothing/interzone_omega.py	c612b36046314f68...25719o	interzone_omega.py — PHASE_XII_SUPRA_CORRIDORS_VEINEUX_Ω_INTERZONE_GENERATION	corridors, zones, salines	—	—
veineux_omega	GOUVERNANCE Ω	ACTIF_Ω	engines/post_smoothing/veineux_omega.py	180b65c30dccb7c6...21073o	veineux_omega.py — PHASE_XII_SUPRA_CORRIDORS_VEINEUX_Ω_ULTIME	corridors	—	—
corridors_vitaux_omega	GOUVERNANCE Ω	ACTIF_Ω	engines/v8_institutional/corridors_vitaux_omega.py	1c6ca2b9e04f88bc...22028o	corridors_vitaux_omega.py — PHASE_XVIII_ENGINE_CORRIDORS_VITAUX_Ω	corridors, zones, salines, hotspots	—	—
ecological_orchestrator_omega	GOUVERNANCE Ω	ACTIF_Ω	engines/v8_institutional/ecological_orchestrator_omega.py	01f9e940a063b6c5...20638o	ecological_orchestrator_omega.py — PHASE_XVII_SUPRA_ENGINE_CORRIDORS_ECOLOGIQUE_Ω	corridors, zones, salines	—	—
engine_affuts	GOUVERNANCE Ω	ACTIF_Ω	engines/v8_institutional/engine_affuts.py	fedc1e8480b48b2b...466o	ENGINE 03 — AFFUTS	—	—	—
engine_audio_acoustique	GOUVERNANCE Ω	ACTIF_Ω	engines/v8_institutional/engine_audio_acoustique.py	f418835be0cea986...4222o	ENGINE 19 — AUDIO-ACOUSTIQUE (sonar passif)	—	—	—
engine_bio_signes	GOUVERNANCE Ω	ACTIF_Ω	engines/v8_institutional/engine_bio_signes.py	734dd78460281c81...4562o	ENGINE 18 — BIO-SIGNES (ADN visuel + chimie visuelle + fraîcheur)	—	—	—
ENGINE-CALIBRATION-DYNAMIQUE-Ω	GOUVERNANCE VIII-X	ACTIF_Ω	engines/v8_institutional/engine_calibration_dynamique_omega.py	e58e46a2b971d286...4720o	ENGINE-CALIBRATION-DYNAMIQUE-Ω — ML ajustement pondérations (Phase X)	zones, hotspots	GET /calibration-dynamique GET /observations POST /observations	—
ENGINE-CALIBRATION-Ω	GOUVERNANCE VIII-X	ACTIF_Ω	engines/v8_institutional/engine_calibration_omega.py	472f4fedd7880fdb...2259o	ENGINE-CALIBRATION-Ω — Calibration non-invasive modeles.	—	—	—

	ENGINE_NAME	PILIER · PHASE	STATUT	PATH	SHA / SIZE	DESCRIPTION FONCTIONNELLE	COUCHES Ω ÉMISES	APIS EXPOSÉES	RISQUES
	ENGINE-CANADA-Ω	GOUVERNANCE Ω	ACTIF_Ω	engines/v8_institutional/engine_canada_omega.py	f8357129a759066a...6004o	ENGINE-CANADA-Ω — Souveraineté pancanadienne (Phase X-B)	corridors	GET /canada GET /canada/province/{code}	—
	engine_carence_nutritionnelle_omega	GOUVERNANCE Ω	ACTIF_Ω	engines/v8_institutional/engine_carence_nutritionnelle_omega.py	7415096d9cae7aeb...2222o	ENGINE_CARENCE_NUTRITIONNELLE_Ω — Détection de carences par espèce.	—	—	—
	engine_champs_nourriciers_omega	GOUVERNANCE Ω	ACTIF_Ω	engines/v8_institutional/engine_champs_nourriciers_omega.py	59508694ba12e9ce...2186o	ENGINE_CHAMPS_NOURRICIERS_Ω — Détection et scoring des champs nourriciers.	—	—	—
	engine_forage_qualite_omega	GOUVERNANCE Ω	ACTIF_Ω	engines/v8_institutional/engine_forage_qualite_omega.py	0a4cf3fbdac8a9f...2326o	ENGINE_FORAGE_QUALITÉ_Ω — Qualité nutritionnelle des végétaux disponibles.	—	—	—
	engine_frequentation	GOUVERNANCE Ω	ACTIF_Ω	engines/v8_institutional/engine_frequentation.py	9b6540714ea346c3...841o	ENGINE 11 — FREQUENTATION (FAUNE + HUMAIN)	—	—	—
	ENGINE-GOUVERNANCE-Ω	GOUVERNANCE Ω	ACTIF_Ω	engines/v8_institutional/engine_gouvernance_omega.py	d96ca368e55a7106...3196o	ENGINE-GOUVERNANCE-Ω — Fusion MONITORING + ALERTE + SCIENCE + REGISTRY.	—	GET /gouvernance	—
	engine_heatmap	GOUVERNANCE Ω	ACTIF_Ω	engines/v8_institutional/engine_heatmap.py	2dd0625a56ebcd03...846o	ENGINE 06 — HEATMAP	—	—	—
	engine_hotspots	GOUVERNANCE Ω	ACTIF_Ω	engines/v8_institutional/engine_hotspots.py	8a268fa092a0499c...1099o	ENGINE 04 — HOTSPOTS	—	—	—
	ENGINE-INCERTITUDE-Ω	GOUVERNANCE Ω	ACTIF_Ω	engines/v8_institutional/engine_incertitude_omega.py	e0210bc0a5f9289a...2866o	ENGINE-INCERTITUDE-Ω — Quantification incertitude par couche/engine/espece.	—	—	—
	engine_intelligence	GOUVERNANCE Ω	ACTIF_Ω	engines/v8_institutional/engine_intelligence.py	7bfce8b9f82359be...1685o	ENGINE 23 — INTELLIGENCE	—	—	—
	ENGINE-NUTRITION-V12-SUPRA	GOUVERNANCE Ω	ACTIF_Ω	engines/v8_institutional/engine_nutrition_v12_supra.py	36fae4fa3e958625...27813o	ENGINE-NUTRITION-V12-SUPRA — Moteur biologique central	—	—	—
	engine_nutritional_attractiveness_omega	GOUVERNANCE Ω	ACTIF_Ω	engines/v8_institutional/engine_nutritional_attractiveness_omega.py	fd7a973154937448...2432o	ENGINE_NUTRITIONAL_ATTRACTIVENESS_Ω — Synthèse attractivité nutritionnelle du territoire.	fusion	—	—
	engine_prediction	GOUVERNANCE Ω	ACTIF_Ω	engines/v8_institutional/engine_prediction.py	7bc14ab5afd386ed...3494o	ENGINE 21 — PREDICTION 48H	—	—	—
	engine_psychologie	GOUVERNANCE Ω	ACTIF_Ω	engines/v8_institutional/engine_psychologie.py	74d0d92a1c34b650...4262o	ENGINE 20 — PSYCHOLOGIE ANIMALE	—	—	—
	ENGINE-QUALIT	GOUVERNANCE Ω	ACTIF_Ω	engines/v8_institutional/engine_qualite_donnees_omega.py	8de74c2a82f4b868...2916o	ENGINE-QUALITÉ-DONNÉES-Ω — Audit complétude/cohérence/fraicheur.	—	—	—
	engine_recettes_salines_omega	GOUVERNANCE Ω	ACTIF_Ω	engines/v8_institutional/engine_recettes_salines_omega.py	b19a0fb23225a101...2112o	ENGINE_RECETTES_SALINES_Ω — Formulations de salines adaptées espèce+carence.	fusion	—	—
	ENGINE-RENDER-Ω	GOUVERNANCE PO-XI-SUPRA	ACTIF_Ω	engines/v8_institutional/engine_render_omega.py	67d7b15a0d6fbe09...6161o	ENGINE-RENDER-Ω — Moteur central de rendu TERRITOIRE (Phase XI-SUPRA)	corridors, zones, affuts, salines, hotspots	POST /render-validate GET /render-config	—
				BCE-4X ULTIME ABSOLU · STEEVE-MAX · CONFORMITÉ Ω 100% · 2026-04-28 RAPPORT_EXHAUSTIF_ENGINES_Ω · 209 engines · 5 piliers · V30 LOCKED INVOLÉ · LECTURE SEULE					
	engine_risque	GOUVERNANCE Ω	ACTIF_Ω	engines/v8_institutional/engine_risque.py	135e6312bd018256...822o	ENGINE 10 — RISQUE	—	—	—
	engine_saisonnalite	GOUVERNANCE Ω	ACTIF_Ω	engines/v8_institutional/engine_saisonnalite.py	07246c83824ef900...1180o	ENGINE 12 — SAISONNALITE	—	—	—
	engine_salines	GOUVERNANCE Ω	ACTIF_Ω	engines/v8_institutional/engine_salines.py	d35f94286f246922...1316o	ENGINE 07 — SALINES	—	—	—
	engine_salines_v11_supra	GOUVERNANCE Ω	ACTIF_Ω	engines/v8_institutional/engine_salines_v11_supra.py	220ff36a3d7b67b6...16176o	ENGINE SALINES-V11-SUPRA — ACTIVATION TOTALE	—	—	—
	engine_sante_physio_omega	GOUVERNANCE Ω	ACTIF_Ω	engines/v8_institutional/engine_sante_physio_omega.py	b3159282414bdd14...1834o	ENGINE_SANTÉ_PHYSIO_Ω — Indicateur de santé physiologique estimée.	—	—	—
	ENGINE-SCIENCE-Ω	GOUVERNANCE Ω	ACTIF_Ω	engines/v8_institutional/engine_science_omega.py	57c4f676eb78e8d8...5272o	ENGINE-SCIENCE-Ω — Registry institutionnel + catalog scientifique	—	—	—
	engine_score_global	GOUVERNANCE Ω	ACTIF_Ω	engines/v8_institutional/engine_score_global.py	92c1bdec76f4888d...8326o	ENGINE 24 — SCORE GLOBAL — MODE REALITE (Phase IX)	zones, hotspots, contamination, sensoriel	—	—
	engine_social_structure_omega	GOUVERNANCE Ω	ACTIF_Ω	engines/v8_institutional/engine_social_structure_omega.py	8e7a125d6a6a470e...1755o	ENGINE_SOCIAL_STRUCTURE_Ω — Structure sociale (grégaire, solitaire, groupe).	—	—	—

	ENGINE_NAME	PILIER · PHASE	STATUT	PATH	SHA / SIZE	DESCRIPTION FONCTIONNELLE	COUCHES Ω ÉMISES	APIs EXPOSÉES	RISQUES
	engine_sol_nutriments_omega	GOUVERNANCE Ω	ACTIF_Ω	engines/v8_institutional/engine_sol_nutriments_omega.py	c36790ed80d054b1... 2472o	ENGINE_SOL_NUTRIMENTS_Ω — Nutriments minéraux et organiques du sol.	—	—	—
	engine_terrain_cost	GOUVERNANCE Ω	ACTIF_Ω	engines/v8_institutional/engine_terrain_cost.py	1cfe791ed033f7f6... 585o	ENGINE 15 — TERRAIN-COST	—	—	—
	engine_trophic_behavior_omega	GOUVERNANCE Ω	ACTIF_Ω	engines/v8_institutional/engine_trophic_behavior_omega.py	3706eb04705385fc... 2555o	ENGINE_TROPHIC_BEHAVIOR_Ω — Comportement trophique par espèce et contexte.	—	—	—
	engine_vent	GOUVERNANCE Ω	ACTIF_Ω	engines/v8_institutional/engine_vent.py	6802864d1f82c1dd... 5181o	ENGINE 05 — VENT	—	—	—
	engine_visibilite	GOUVERNANCE Ω	ACTIF_Ω	engines/v8_institutional/engine_visibilite.py	03b87375a10dac2d... 791o	ENGINE 16 — VISIBILITE	—	—	—
	engine_zones	GOUVERNANCE Ω	ACTIF_Ω	engines/v8_institutional/engine_zones.py	8229ca7c0d16e5f6... 503o	ENGINE 01 — ZONES	—	—	—
	esi_omega	GOUVERNANCE Ω	ACTIF_Ω	engines/v8_institutional/esi_omega.py	bed38dd357709e35... 18261o	ENGINE SECURITE-INSTITUTIONNELLE: Omega (ESI-Omega)	corridors, zones, affuts, salines, hotspots	GET /laws GET /conformite/full GET /status	—
	export_institutionnel_v20_omega	GOUVERNANCE PO-XI-SUPRA	ACTIF_Ω	engines/v8_institutional/export_institutionnel_v20_omega.py	7f6442074fb0ed12... 6954o	EXPORT-INSTITUTIONNEL-V20-Ω — PDF signé (Phase X-D)	—	GET /export/institutionnel/v20	—
	federal_datasets_omega	GOUVERNANCE Ω	ACTIF_Ω	engines/v8_institutional/federal_datasets_omega.py	a33be806485a4cce... 6974o	FEDERAL-DATASETS-Ω — LEP + HYDAT (Phase X-C)	—	GET /federal/hydat/province/{code} GET /federal/lep GET /federal/hydat	—
	fusion_territoire_omega	GOUVERNANCE Ω	ACTIF_Ω	engines/v8_institutional/fusion_territoire_omega.py	3c1c33056028011e... 20106o	fusion_territoire_omega.py — PHASE-E DOCTRINE PERMANENTE 50% + DÉROGATION BIO	contamination, score_ultime	—	—
	lep_ingestion_omega	GOUVERNANCE Ω	ACTIF_Ω	engines/v8_institutional/lep_ingestion_omega.py	67549a5350b52264... 13152o	LEP-INGESTION-Ω — BIONIC INGESTION-FGDB+GEOJSON-Ω-V1.0 (Phase XI-SUPRA-D)	—	POST /ingest GET /registry GET /geojson/{layer}	—
	lidar_irda_v11	GOUVERNANCE Ω	ACTIF_Ω	engines/v8_institutional/lidar_irda_v11.py	a8d9b85c636530b5... 18041o	ENGINE LIDAR-IRDA V11 — INGESTION DONNÉES INSTITUTIONNELLES 1m	—	—	—
	ENGINE-MONITORING-Ω	GOUVERNANCE PO-XI-SUPRA	ACTIF_Ω	engines/v8_institutional/monitoring_alerte_omega.py	c8ad5dd19e9d9184... 4424o	ENGINE-MONITORING-Ω + ENGINE-ALERTE-ANOMALIES-Ω	—	GET /monitoring GET /alertes	—
	origine_externe_filter_omega	GOUVERNANCE Ω	ACTIF_Ω	engines/v8_institutional/origine_externe_filter_omega.py	33706d4353a5e149... 13388o	origine_externe_filter_omega.py — PHASE_XIX_P1_ORIGINE_EXTERNE_FILTER_Ω	corridors	—	—
	origine_externe_inversion_omega	GOUVERNANCE Ω	ACTIF_Ω	engines/v8_institutional/origine_externe_inversion_omega.py	2f0ae33a99c312f9... 9079o	origine_externe_inversion_omega.py — PHASE_XIX_P2_ORIGINE_EXTERNE_INVERSION_Ω	corridors	—	—
	predictive_omega_v2	GOUVERNANCE Ω	ACTIF_Ω	engines/v8_institutional/predictive_omega_v2.py	95d10ddcf2a63cf2... 16715o	predictive_omega_v2.py — PHASE_XVIII_ENGINE_PREDICTIVE_OMEGA_GPS_USGS_Ω	corridors	—	—
	protections_omega	GOUVERNANCE Ω	ACTIF_Ω	engines/v8_institutional/protections_omega.py	51fc00d06405fcb4... 3772o	protections_omega.py — PHASE 2 STABILISATION TERRITOIRE Ω	—	—	—
	redis_omega	GOUVERNANCE Ω	ACTIF_Ω	engines/v8_institutional/redis_omega.py	f8b2ea439ca40fcc... 4717o	REDIS-Ω — Backend cache partage multi-pod	—	—	—
	ENGINE-SCIENCE-Ω	GOUVERNANCE VIII-X	ACTIF_Ω	engines/v8_institutional/registry_lock_omega.py	fb765b94cc1fd421... 8703o	REGISTRY-LOCK-Ω — Verrouillage institutionnel Phase XI	—	GET /document-maitre-lock GET /registry-lock	—
	science_gaps_datasets	GOUVERNANCE Ω	ACTIF_Ω	engines/v8_institutional/science_gaps_datasets.py	3df60c3f518cf122... 5475o	SCIENCE-Ω GAPS DATASETS — Seed institutionnels QC (Phase X)	zones	GET /science-gaps	—
	ENGINE-SHIELD-0	GOUVERNANCE Ω	ACTIF_Ω	engines/v8_institutional/securite_omega_v19.py	4efb2907b1444fe3... 10811o	SECURITE-Omega V19 — SHIELDS + GUARDS + VALIDATORS	corridors, zones, salines, contamination	—	—
	self_audit_alerts_omega	GOUVERNANCE VIII-X	ACTIF_Ω	engines/v8_institutional/self_audit_alerts_omega.py	090df97bbe9011d1... 4968o	SELF-AUDIT-ALERTS-Ω — Canal WebSocket institutionnel (Phase X-D)	—	—	—
	self_audit_omega	GOUVERNANCE VIII-X	ACTIF_Ω	engines/v8_institutional/self_audit_omega.py	449b6d0fe48c53a8... 14550o	SELF-AUDIT-Ω — Validation institutionnelle TERRITOIRE-V12	—	GET /self-audit GET /self-audit/last	—

ENGINE_NAME	PILIER · PHASE	STATUT	PATH	SHA / SIZE	DESCRIPTION FONCTIONNELLE	COUCHES Ω ÉMISES	APIS EXPOSÉES	RISQUES
sla_baseline_30j_omega	GOUVERNANCE P0-XI-SUPRA	ACTIF_Ω	engines/v8_institutional/sla_baseline_30j_omega.py	dcc4621c4400cdb5... 3700o	SLA-BASELINE-30J-Ω — Série temporelle 30 jours (Phase X-D)	—	GET /sla-baseline-30j	—
sla_baseline_omega	GOUVERNANCE P0-XI-SUPRA	ACTIF_Ω	engines/v8_institutional/sla_baseline_omega.py	c3033785bc40c2a8... 16829o	SLA-BASELINE-Ω — Baseline institutionnelle + PERF-GUARD-Ω	corridors	GET /sla-baseline POST /sla-baseline/seed DELETE /sla-baseline	—
ENGINE-SUPRA-DONNEES-0	GOUVERNANCE Ω	ACTIF_Ω	engines/v8_institutional/supra_donnees.py	04a81ec1369718b8... 14396o	ENGINE SUPRA-DONNEES-Omega — CENTRAL DATA INTELLIGENCE	—	—	—
terrain_v10_supra	GOUVERNANCE Ω	ACTIF_Ω	engines/v8_institutional/terrain_v10_supra.py	61e38e5a8f813dad... 15619o	ENGINE DONNEES-REELLES-TERRAIN V10-SUPRA-INSTITUTIONNEL	—	—	—
ENGINE-TERRAIN-V10-SUPRA	GOUVERNANCE Ω	ACTIF_Ω	engines/v8_institutional/territoire_v10_supra.py	06601e0040e30e95... 66444o	TERRITOIRE V10-SUPRA — FUSION TOTALE REEL + IA	corridors, zones, affuts, salines, hotspots, contamination	—	—
v20_performance_bundle	GOUVERNANCE Ω	ACTIF_Ω	engines/v8_institutional/v20_performance_bundle.py	5b55efb17b53fa9b... 21920o	V20 PERFORMANCE BUNDLE — V11-SUPRA SCALABILITE 10K	corridors, zones, affuts, contamination	POST /bundle/purge GET /bundle/stats POST /bundle/warmup	—
visual_proof_live_omega	GOUVERNANCE P0-XI-SUPRA	ACTIF_Ω	engines/v8_institutional/visual_proof_live_omega.py	6ef5047c7efe5099... 7182o	VISUAL-PROOF-LIVE-Ω — Capture DOM Leaflet réelle (Phase XI-SUPRA-C)	—	GET /visual-proof-live/index POST /visual-proof-live/generate	—
ENGINE-RENDER-Ω	GOUVERNANCE P0-XI-SUPRA	ACTIF_Ω	engines/v8_institutional/visual_proof_omega.py	c14cdcd5a94d0caa... 13671o	VISUAL-PROOF-Ω — Preuve visuelle institutionnelle (Phase XI-SUPRA-B)	corridors, affuts, salines, hotspots	GET /visual-proof/index POST /visual-proof/generate	—
ENGINE-V8	GOUVERNANCE X-B	ACTIF_Ω	engines/v8_national/exclusion_engine.py	42cbcd5d1c99066b... 21373o	EXCLUSION-ENGINE-V8 — Moteur centralise d'exclusion BCE-4X	contamination	GET /decision GET /status GET /referential	—
governance	GOUVERNANCE X-B	ACTIF_Ω	engines/v8_national/governance.py	690ff6835b7eea1b... 7661o	V8-GOVERNANCE — Master Switch Admin Premium SUPREMACY	—	GET /audit POST /activate GET /state	—
map_bundle	GOUVERNANCE X-B	ACTIF_Ω	engines/v8_national/map_bundle.py	05329cc8c3d2d219... 13480o	V8-MAP-BUNDLE — Endpoint unique toutes couches territoire	corridors, zones, affuts	GET /bundle GET /bundle/status	—
ENGINE-V7	GOUVERNANCE X-B	ACTIF_Ω	engines/v8_national/p1_pipelines.py	ba2d682e255481db... 10454o	V8-P1 — Pipelines donnees reelles (LIDAR WCS + IRDA Pedologie)	—	GET /lidar GET /status GET /pedology	—
phase_a_engines	GOUVERNANCE X-B	ACTIF_Ω	engines/v8_national/phase_a_engines.py	32f454152ef7dea2... 16232o	V8-REINTEGRATION-PHASE-A — RELOCALISATION + SALINES	salines, contamination	GET /relocalisation GET /salines GET /phase-a/status	—
phase_b_engines	GOUVERNANCE X-B	ACTIF_Ω	engines/v8_national/phase_b_engines.py	ede994e06f35459e... 33451o	V8-REINTEGRATION-PHASE-B — ZONES + CORRIDORS + AFFUTS TERRAIN-AWARE	corridors, zones, affuts	GET /affuts-ta GET /zones-ta GET /corridors-ta	—
phase_c_engines	GOUVERNANCE X-B	ACTIF_Ω	engines/v8_national/phase_c_engines.py	51746ac46e3b629a... 15424o	V8-PHASE-C — SCENARIO + THERMAL + MULTI-ENGINE SCORING	corridors, zones, affuts	GET /scenario/presets GET /multi-score GET /phase-c/status	—
referentials	GOUVERNANCE X-B	ACTIF_Ω	engines/v8_national/referentials.py	0c610c3466b853a6... 13401o	V8-NATIONAL — Referentiels nationaux pancanadiens	—	—	—
engine_corridors_legacy_pre_L	GOUVERNANCE Ω	INACTIF_V8_INST	engines/v8_institutional/_ARCHIVE_NON_ACTIVE/engine_corridors_legacy_pre_L.py	db78bb0e1bdd5ada... 640o	ENGINE 02 — CORRIDORS V9-x20	—	—	Orphelin V8_INST — validation contextuell

ENGINE_NAME	PILIER · PHASE	STATUT	PATH	SHA / SIZE	DESCRIPTION FONCTIONNELLE	COUCHES Ω ÉMISES	APIs EXPOSÉES	RISQUES
__init__	GOUVERNANCE Ω	INACTIF_V8_INST	engines/v8_institutional/__init__.py	560692b53e3ee83b... 305o	BIONIC OS — V8-ENGINES-INSTITUTIONNEL-Ω-ULTIME-MAX	—	—	Orphelin V8_INST — validation contextuell
engine_cameras	GOUVERNANCE Ω	INACTIF_V8_INST	engines/v8_institutional/engine_cameras.py	4c2a75ffa1c8b5ee... 362o	ENGINE 17 — CAMERAS	—	—	Orphelin V8_INST — validation contextuell
ENGINE-IA-VISION-REGISTRY-Ω	GOUVERNANCE Ω	INACTIF_V8_INST	engines/v8_institutional/engine_ia_vision_registry_omega.py	7abd72004637d8bb... 3482o	ENGINE-IA-VISION-REGISTRY-Ω — Registre préparatoire IA Vision	—	GET /validate GET /status	Orphelin V8_INST — validation contextuell
engine_nutrition	GOUVERNANCE Ω	INACTIF_V8_INST	engines/v8_institutional/engine_nutrition.py	04110915819d435e... 905o	ENGINE 08 — NUTRITION-MINERAUX	—	—	Orphelin V8_INST — validation contextuell
ENGINE-RENDU-Ω	GOUVERNANCE P0-XI-SUPRA	INACTIF_V8_INST	engines/v8_institutional/engine_rendu_omega.py	96af50ad96bb7b6b... 13029o	ENGINE-RENDU-Ω — Moteur de rendu institutionnel des corridors (Phase XI-SUPRA-K)	corridors, zones, affuts, salines, hotspots	POST /validate GET /status GET /rules	Orphelin V8_INST — validation contextuell
ENGINE-TERRITOIRE-ANTI-REGRESSION-Ω	GOUVERNANCE P0-XI-SUPRA	INACTIF_V8_INST	engines/v8_institutional/engine_territoire_anti_regression_omega.py	b7ddafc36500ca6d... 14013o	ENGINE-TERRITOIRE-ANTI-REGRESSION-Ω — Phase XI-SUPRA-G (ORDRE PROTECT-Ω)	corridors, zones, affuts, hotspots, contamination	POST /validate POST /seal-baseline GET /status	Orphelin V8_INST — validation contextuell
engines_catalog	GOUVERNANCE Ω	INACTIF_V8_INST	engines/v8_institutional/engines_catalog.py	5687cf6bb94815ed... 1358o	ENGINES-CATALOG-Ω — Endpoint gouvernance	—	GET /engines-catalog	Orphelin V8_INST — validation contextuell
hotspots_organic_v1	GOUVERNANCE Ω	INACTIF_V8_INST	engines/v8_institutional/hotspots_organic_v1.py	1b8b074690089365... 2152o	STUB — hotspots_organic_v1 (Phase XI-SUPRA-M PREP — NON-Ω)	—	—	Orphelin V8_INST — validation contextuell
phase_omega_secure_lockdown	GOUVERNANCE Ω	INACTIF_V8_INST	engines/v8_institutional/phase_omega_secure_lockdown.py	198dd8a9ff5dd97a... 18370o	PHASE_Ω-SECURE-REACTIVATION — FULL_STACK_LOCKDOWN_V12 — CORRIDORS_Ω-CERTIFICATION	corridors, zones, affuts, salines, hotspots	—	Orphelin V8_INST — validation contextuell
ENGINE-IA-CORRIDORS-ORGANIC-Ω	GOUVERNANCE Ω	INACTIF_V8_INST	engines/v8_institutional/piliers_router.py	c9f6da743d51a3ec... 15666o	4 PILIERS STRUCTURELS — ORCHESTRATEURS MAITRES	corridors, zones, affuts, salines, hotspots, contamination	GET /pilier/prediction-intelligence GET /pilier/systeme-sensoriel GET /pilier/comportement-humain	Orphelin V8_INST — validation contextuell
salines_organic_v1	GOUVERNANCE Ω	INACTIF_V8_INST	engines/v8_institutional/salines_organic_v1.py	2956ed8ba048e22e... 2182o	STUB — salines_organic_v1 (Phase XI-SUPRA-M PREP — NON-Ω)	—	—	Orphelin V8_INST — validation contextuell
supra_v8	GOUVERNANCE Ω	INACTIF_V8_INST	engines/v8_institutional/supra_v8.py	519a9b3fc5a4ee4d... 7974o	SUPRA V8 — Integration TERRITOIRE → SUPRA	—	GET /recommandation GET /score GET /analyse	Orphelin V8_INST — validation contextuell
v20_mvt_tiles	GOUVERNANCE Ω	INACTIF_V8_INST	engines/v8_institutional/v20_mvt_tiles.py	57deb24e20a17fd5... 20093o	V20 MVT TILES — PHASE-PERFORMANCE-Omega V11-SUPRA	corridors, zones, affuts, salines, hotspots, contamination	GET /stats POST /purge GET /{layer}/{z}/{x}/{y}.json	Orphelin V8_INST — validation contextuell

ENGINE_NAME	PILIER · PHASE	STATUT	PATH	SHA / SIZE	DESCRIPTION FONCTIONNELLE	COUCHES Ω ÉMISES	APIs EXPOSÉES	RISQUES
visual_proof_live_playwright	GOUVERNANCE P0-XI-SUPRA	INACTIF_V8_INST	engines/v8_institutional/visual_proof_live_playwright.py	491138c341391a36...10881o	Playwright capture script — Phase XI-SUPRA-D	—	—	Orphelin V8_INST — validation contextuell
zones_organic_v1	GOUVERNANCE Ω	INACTIF_V8_INST	engines/v8_institutional/zones_organic_v1.py	87077dae87f3c204...2201o	STUB — zones_organic_v1 (Phase XI-SUPRA-M PREP — NON-Ω)	—	—	Orphelin V8_INST — validation contextuell
__init__	GOUVERNANCE Ω-X	INACTIF_ORPHELIN	engines/legal_time_omega/__init__.py	5724b7d5ed8e894a...532o	ENGINE_LEGAL_TIME_Ω — Package squelette X199-PREPARATOIRE	—	—	—
__init__	GOUVERNANCE X-B	INACTIF_ORPHELIN	engines/v8_national/__init__.py	345acca0100309a5...52o		—	—	—
ENGINE-V8	GOUVERNANCE X-B	INACTIF_ORPHELIN	engines/v8_national/router.py	f18a2e5b35a832da...13203o	V8-NATIONAL — Router API moteurs nationaux	corridors, zones, affuts	GET /species-profile GET /score GET /status	—
router	BIO-SYSTEME Ω-P1	ACTIF_Ω	engines/bio_scoring_omega/router.py	690b9b8d90286b85...3768o	ENGINE_BIO_SCORING_Ω — Implémentation X200 P0 (support)	—	GET /status POST /compute	—
v30_mirror_read_only	BIO-SYSTEME Ω-P1	ACTIF_Ω	engines/bio_scoring_omega/v30_mirror_read_only.py	45d7d0ceebad90bf...5222o	v30_mirror_read_only.py — Façade miroir lecture seule V30	—	—	—
router	BIO-SYSTEME Ω-P3	ACTIF_Ω	engines/predictive_omega/router.py	09b15150fc1ba4e3...6275o	ENGINE_PREDICTIVE_Ω — Router FastAPI — X199 ACTIVÉ	—	POST / GET /status	—
external_inflow	BIO-SYSTEME Ω-XII	ACTIF_Ω	engines/reseau_veineux_omega/external_inflow.py	bfd70e6944aae817...16534o	external_inflow.py — Logique EXTERNAL INFLOW pour ENGINE RÉSEAU VEINEUX_Ω	—	—	—
router	BIO-SYSTEME Ω-XII	ACTIF_Ω	engines/reseau_veineux_omega/router.py	0e2eb7cbcd7eec7a...13401o	ENGINE_RESEAU_VEINEUX_Ω — Implémentation X200 P0 (support)	salines, fusion	POST /external-inflow/preview GET /external-inflow/status GET /status	—
engine_connectivite	BIO-SYSTEME P1-XI	ACTIF_Ω	engines/v8_institutional/engine_connectivite.py	a36987bcbdef1771...737o	ENGINE 22 — CONNECTIVITE	—	—	—
ENGINE-CONNECTIVITE-ECOLOGIQUE-Ω	BIO-SYSTEME P1-XI	ACTIF_Ω	engines/v8_institutional/engine_connectivite_ecologique_omega.py	38f81a039dd91918...1248o	ENGINE-CONNECTIVITE-ECOLOGIQUE-Ω — Connectivité corridors + evitement routes.	—	—	—
ENGINE-CONTAMINATION-Ω	BIO-SYSTEME P1-XI	ACTIF_Ω	engines/v8_institutional/engine_contamination_v2_omega.py	f2edde81070818a5...4313o	ENGINE-CONTAMINATION-Ω V2 + MODULE ANTI-CONTAMINATION-INSTITUTIONNEL-Ω	—	—	—
ENGINE-ESPECE-Ω	BIO-SYSTEME P1-XI	ACTIF_Ω	engines/v8_institutional/engine_espece_omega.py	ec260ce8218ca021...975o	ENGINE-ESPECE-Ω — Profils espèces depuis catalog SCIENCE-Ω.	—	—	—
ENGINE-HABITAT-SUPRA	BIO-SYSTEME P1-XI	ACTIF_Ω	engines/v8_institutional/engine_habitat_supra.py	4a39bfdc21f375f1...4631o	ENGINE-HABITAT-SUPRA — Moteur habitat institutionnel	—	—	—
ENGINE-IA-CORRIDORS-Ω	BIO-SYSTEME P1-XI	ACTIF_Ω	engines/v8_institutional/engine_ia_corridors_omega.py	bcb1e3a6a92304a1...17448o	ENGINE IA-CORRIDORS-Ω — Phase XI-SUPRA-H	corridors, salines	POST /validate POST /validate-live GET /status	—
ENGINE-IA-CORRIDORS-ORGANIC-Ω	BIO-SYSTEME P1-XI	ACTIF_Ω	engines/v8_institutional/engine_ia_corridors_organic_omega.py	027712696407882f...55197o	ENGINE-IA-CORRIDORS-ORGANIC-Ω — Phase XI-SUPRA-M (CORRIDORS_ORGANIC_OMEGA)	corridors, zones, salines, hotspots, contamination	POST /validate POST /seal-baseline POST /generate-alt	—
ENGINE-IA-VISION-ECOLOGIQUE-Ω	BIO-SYSTEME P1-XI	ACTIF_Ω	engines/v8_institutional/engine_ia_vision_ecologique_omega.py	40204496b3a08575...1584o	ENGINE-IA-VISION-ECOLOGIQUE-Ω — IA Vision structure foret + zones probables.	—	—	—
engine_optimisation_habitat_omega	BIO-SYSTEME P1-XI	ACTIF_Ω	engines/v8_institutional/engine_optimisation_habitat_omega.py	2710f9ef696293fb...3143o	ENGINE_OPTIMISATION_HABITAT_Ω — Synthèse habitat optimisé (territoire ULTIME).	fusion	—	—
ENGINE-POPULATION-DYNAMICS-Ω	BIO-SYSTEME P1-XI	ACTIF_Ω	engines/v8_institutional/engine_population_dynamics_omega.py	dfde1cedf8653ff8...4463o	ENGINE-POPULATION-DYNAMICS-Ω — Demographie multi-especes.	—	—	—
ENGINE-SPECIES-PROFILES-Ω	BIO-SYSTEME P1-XI	ACTIF_Ω	engines/v8_institutional/engine_species_profiles_omega.py	635f9bf004c388f4...4755o	ENGINE-SPECIES-PROFILES-Ω — Registre dynamique des profils d'espèces	—	GET /validate GET /status GET /{species_key}	—

	ENGINE_NAME	PILIER · PHASE	STATUT	PATH	SHA / SIZE	DESCRIPTION FONCTIONNELLE	COUCHES Ω ÉMISES	APIS EXPOSÉES	RISQUES
	species_modulator_omega	BIO-SYSTEME P1-XI	ACTIF_Ω	engines/v8_institutional/species_modulator_omega.py	694ec91834184c57... 9233o	species_modulator_omega.py — PHASE_XVI_ENGINE_CORRIDORS_UNIFIÉ_Ω	—	—	—
	species_presence_mask_omega	BIO-SYSTEME P1-XI	ACTIF_Ω	engines/v8_institutional/species_presence_mask_omega.py	9a0b42a9f2ea097c... 15218o	species_presence_mask_omega.py — PHASE_XVIII_BIO_PRESENCE_MASK_Ω	corridors, zones, affuts, salines, hotspots, contamination	—	—
	species_weighting_profiles	BIO-SYSTEME P1-XI	ACTIF_Ω	engines/v8_institutional/species_weighting_profiles.py	fd291adaccaa3a07... 3460o	SPECIES_WEIGHTING_PROFILES — variantes pondérations par espèce (Phase X).	zones, hotspots, sensoriel	—	—
	router	BIO-SYSTEME Ω-P1	ACTIF_Ω	engines/wildlife_behavior_omega/router.py	bf35ea0b0e59ca28... 4910o	ENGINE_WILDLIFE_BEHAVIOR_Ω — Implémentation X200 P0	—	GET /status POST /compute	—
	__init__	BIO-SYSTEME Ω-P1	INACTIF_ORPHELIN	engines/bio_scoring_omega/__init__.py	941fa2344776dbf... 520o	ENGINE_BIO_SCORING_Ω — Package squelette X199-PREPARATOIRE	—	—	—
	__init__	BIO-SYSTEME Ω-P3	INACTIF_ORPHELIN	engines/predictive_omega/__init__.py	95c1a1fbf60c81c3... 542o	ENGINE_PREDICTIVE_Ω — Package squelette X199-PREPARATOIRE	—	—	—
	__init__	BIO-SYSTEME Ω-XII	INACTIF_ORPHELIN	engines/reseau_veineux_omega/__init__.py	17954cdf7161b634... 537o	ENGINE_RESEAU_VEINEUX_Ω — Package squelette X199- PREPARATOIRE	—	—	—
	__init__	BIO-SYSTEME Ω-P1	INACTIF_ORPHELIN	engines/wildlife_behavior_omega/__init__.py	8424303e2ced6375... 551o	ENGINE_WILDLIFE_BEHAVIOR_Ω — Package squelette X199- PREPARATOIRE	—	—	—
	engine_comportement	COMPORTEMENT- HUMAIN P1-SUPRA	ACTIF_Ω	engines/v8_institutional/engine_comportement.py	7b17df3676ccb1f4... 1195o	ENGINE_13 — COMPORTEMENT	—	—	—
	engine_comportement_avance	COMPORTEMENT- HUMAIN P1-SUPRA	ACTIF_Ω	engines/v8_institutional/engine_comportement_avance.py	6bed1da8553ed6d5... 681o	ENGINE_14 — COMPORTEMENT-AVANCE	—	—	—
	ENGINE-COMPORTEMENT-BIOLOGIQUE-Ω	COMPORTEMENT- HUMAIN P1-SUPRA	ACTIF_Ω	engines/v8_institutional/engine_comportement_biologique_omega.py	7278489084cbbc28... 1922o	ENGINE-COMPORTEMENT-BIOLOGIQUE-Ω — Patterns comportementaux saisonniers.	corridors	—	—
	ENGINE-STRESS-ANTHROPIQUE-Ω	COMPORTEMENT- HUMAIN P1-SUPRA	ACTIF_Ω	engines/v8_institutional/engine_stress_anthropique_omega.py	b1796c5da86fd8f2... 4102o	ENGINE-STRESS-ANTHROPIQUE-Ω — Moteur pression humaine institutionnel	—	—	—
	ENGINE-SENSORIEL-VENT-ODEURS-Ω	SYSTEME- SENSORIEL P1	ACTIF_Ω	engines/v8_institutional/engine_sensoriel_vent_odeurs_omega.py	cd0d07a06debd8d9... 1768o	ENGINE-SENSORIEL-VENT-ODEURS-Ω — Dispersion olfactive + detection vent.	—	—	—
	router	ENVIRONNEMENT Ω-SUPRA	ACTIF_Ω	engines/advanced_geospatial_omega/router.py	1ba26c9c0c7874a4... 6344o	ENGINE_ADVANCED_GEOSPATIAL_Ω — Router FastAPI — X199 ACTIVÉ	fusion	POST / GET /status	—
	router	ENVIRONNEMENT Ω-SUPRA	ACTIF_Ω	engines/eco_zones_omega/router.py	edc97ff665dd230c4... 6403o	ENGINE_ECO_ZONES_Ω — Implémentation X200 P0	zones, salines	GET /status POST /compute GET /saline-sources	—
	router	ENVIRONNEMENT Ω SUPRA	ACTIF_Ω	engines/ecoforestry_omega/router.py	37f26422cd92d62a... 5765o	ENGINE_ECOFORESTRY_Ω — Router FastAPI — X199 ACTIVÉ	—	POST / GET /status	—
	router	ENVIRONNEMENT Ω-SUPRA	ACTIF_Ω	engines/hydro_topo_omega/router.py	64d5f7f4eb8b0619... 4611o	ENGINE_HYDRO_TOPO_Ω — Implémentation X200 P0	—	GET /status POST /compute	—
	router	ENVIRONNEMENT Ω-SUPRA	ACTIF_Ω	engines/terrain_3d_omega/router.py	1e25c08a5fb6c5af... 5267o	ENGINE_3D_TERRAIN_Ω — Router FastAPI — X199 ACTIVÉ	—	POST / GET /status	—
	engine_canopee_thermique_omega	ENVIRONNEMENT P1-X-C	ACTIF_Ω	engines/v8_institutional/engine_canopee_thermique_omega.py	bab5b1059f5cd3d0... 1352o	ENGINE_CANOPEE_THERMIQUE_Ω — Effet thermique canopée (ombre/refuge).	—	—	—
	ENGINE-CLIMAT-FUTUR-Ω	ENVIRONNEMENT P1-X-C	ACTIF_Ω	engines/v8_institutional/engine_climat_futur_omega.py	bc7613fe2a5d805f... 2679o	ENGINE-CLIMAT-FUTUR-Ω — Projections CMIP6 calibrees.	—	—	—
	ENGINE-HYDROLOGIE-SUPRA	ENVIRONNEMENT P1-X-C	ACTIF_Ω	engines/v8_institutional/engine_hydrologie_supra.py	161f4c611d0c2521... 2771o	ENGINE-HYDROLOGIE-SUPRA — Moteur hydrologie institutionnel	—	—	—
	ENGINE-INFLUENCE-LUNAIRE-Ω	ENVIRONNEMENT P1-X-C	ACTIF_Ω	engines/v8_institutional/engine_influence_lunaire_omega.py	ff67806c93cc8446... 2716o	ENGINE-INFLUENCE-LUNAIRE-Ω — Phase lunaire + activite faunique.	—	—	—

ENGINE_NAME	PILIER · PHASE	STATUT	PATH	SHA / SIZE	DESCRIPTION FONCTIONNELLE	COUCHES Ω ÉMISES	APIs EXPOSÉES	RISQUES
engine_microclimat_advanced_omega	ENVIRONNEMENT P1-X-C	ACTIF_Ω	engines/v8_institutional/engine_microclimat_advanced_omega.py	3f6724c18d0facca… 2113o	ENGINE_MICROCLIMAT_Ω_ADVANCED — Version avancée microclimat locale.	—	—	—
engine_pression	ENVIRONNEMENT P1-X-C	ACTIF_Ω	engines/v8_institutional/engine_pression.py	db4c96b85fd3c844… 763o	ENGINE 09 — PRESSION HUMAINE	—	—	—
ENGINE-PRESSION-ATMOSPHERIQUE-Ω	ENVIRONNEMENT P1-X-C	ACTIF_Ω	engines/v8_institutional/engine_pression_atmospherique_omega.py	5b839db0039707b7… 2446o	ENGINE-PRESSION-ATMOSPHERIQUE-Ω — Pression + tendance + impact faunique.	—	—	—
ENGINE-RISQUES-HYDRO-Ω	ENVIRONNEMENT P1-X-C	ACTIF_Ω	engines/v8_institutional/engine_risques_hydro_omega.py	5b14858da230d752… 1799o	ENGINE-RISQUES-HYDRO-Ω — Risques hydrologiques pancanadiens (Phase X-C)	—	GET /risques-hydro	—
ENGINE-SOL-SUPRA	ENVIRONNEMENT P1-X-C	ACTIF_Ω	engines/v8_institutional/engine_sol_supra.py	e68328855768dae2… 2884o	ENGINE-SOL-SUPRA — Moteur pédologie institutionnel	—	—	—
ENGINE-THERMIQUE-MICROCLIMAT-Ω	ENVIRONNEMENT P1-X-C	ACTIF_Ω	engines/v8_institutional/engine_thermique_microclimat_omega.py	89b85680328ec47e… 1570o	ENGINE-THERMIQUE-MICROCLIMAT-Ω — Stress thermique + microclimat.	—	—	—
__init__	ENVIRONNEMENT Ω-SUPRA	INACTIF_ORPHELIN	engines/advanced_geospatial_omega/__init__.py	c9d83e77b014e5dc… 538o	ENGINE_ADVANCED_GEOSPATIAL_Ω — Package squelette X199-PRÉPARATOIRE	—	—	—
__init__	ENVIRONNEMENT Ω-SUPRA	INACTIF_ORPHELIN	engines/eco_zones_omega/__init__.py	f5e350081f453163… 522o	ENGINE_ECO_ZONES_Ω — Package squelette X199-PRÉPARATOIRE	—	—	—
__init__	ENVIRONNEMENT Ω-SUPRA	INACTIF_ORPHELIN	engines/ecoforestry_omega/__init__.py	4f3ab419d74e3f11… 525o	ENGINE_ECOFORESTRY_Ω — Package squelette X199-PRÉPARATOIRE	—	—	—
__init__	ENVIRONNEMENT Ω-SUPRA	INACTIF_ORPHELIN	engines/hydro_topo_omega/__init__.py	bdeef00259211b32… 529o	ENGINE_HYDRO_TOPO_Ω — Package squelette X199-PRÉPARATOIRE	—	—	—
__init__	ENVIRONNEMENT Ω-SUPRA	INACTIF_ORPHELIN	engines/terrain_3d_omega/__init__.py	5f25b11a7171ecb8… 508o	ENGINE_3D_TERRAIN_Ω — Package squelette X199-PRÉPARATOIRE	—	—	—
ENGINE-IA-CORRIDORS-ORGANIC-Ω	LEGACY ?	ACTIF_Ω	engines/post_smoothing/organic_corridor_smoother.py	7cf53689fb7f42b7… 40519o	organic_corridor_smoother.py — Post-processeur RENDU Ω externe	corridors, salines	GET /smoother-status POST /generate	—
p1_preparation	LEGACY ?	ACTIF_Ω	engines/post_smoothing/p1_preparation.py	0144ca6155b784cb… 24504o	p1_preparation.py — Préparation P1 & Branchement P1.2 EXTERNAL INFLOW → SMOOTHER X180	corridors, salines, fusion	—	—
predictive_integration	LEGACY ?	ACTIF_Ω	engines/post_smoothing/predictive_integration.py	47dfb202ea545ea4… 11632o	predictive_integration.py — Intégration ENGINE_PREDICTIVE_Ω dans smoother X180	corridors	—	—
renduomega	LEGACY ?	ACTIF_Ω	engines/post_smoothing/renduomega.py	1c14ec3edf789e46… 28821o	ENGINE_RENDUΩ — Validation stricte des corridors avant publication	corridors, zones, affuts, salines, hotspots, contamination	—	—
terrain_signals_builder	LEGACY ?	ACTIF_Ω	engines/post_smoothing/terrain_signals_builder.py	a110fc3d36e9b053… 15360o	terrain_signals_builder.py — Générateur de terrain_signals institutionnels Ω	—	—	—
x199_commons	LEGACY ?	ACTIF_Ω	engines/x199_commons.py	e916dbc9527b6f19… 2017o	x199_commons.py — Triple verrou Ω commun aux 5 engines X199	—	—	—
anomaly_detector	LEGACY ?	ACTIF_LEGACY	engines/bdre/anomaly_detector.py	396741ae948b2e5e… 5367o	BDRE — Anomaly Detector (F3)	—	—	—
corridor_optimizer_v2	LEGACY ?	ACTIF_LEGACY	engines/bdre/corridor_optimizer_v2.py	f8dedd4109df8cc5… 23822o	corridor_optimizer_v2.py — CORRIDOR-FIRST X1 000 000%	—	—	—
health_monitor	LEGACY ?	ACTIF_LEGACY	engines/bdre/health_monitor.py	b0ef40b124744c08… 2836o	BDRE — Health Monitor (F1)	—	—	—
institutional_cache	LEGACY ?	ACTIF_LEGACY	engines/bdre/institutional_cache.py	af61b52c7cec6261… 15430o	BCE-4X — CACHE INSTITUTIONNEL PERMANENT	affuts	—	—
corridor_builder	LEGACY ?	ACTIF_LEGACY	engines/corridor_unified/corridor_builder.py	f4e4f98d4e5b483c… 9557o	BCE-4X BLOC 1 — CORRIDOR_UNIFIED BUILDER	—	—	—
corridor_model	LEGACY ?	ACTIF_LEGACY	engines/corridor_unified/corridor_model.py	bd85c47997f3a690… 9254o	BCE-4X BLOC 1 — CORRIDOR_UNIFIED MODEL	—	—	—
access_engine	LEGACY ?	ACTIF_LEGACY	engines/hunt_orchestrator/access_engine.py	1f6d82167783d9ca… 35599o	BCE-4X PO — ENGINE ACCES DYNAMIQUE v2	—	—	—
choix_affuts	LEGACY ?	ACTIF_LEGACY	engines/hunt_orchestrator/choix_affuts.py	846af306006ce60a… 16874o	BCE-4X PO-C — ENGINE CHOIX DES AFFUTS v2	—	—	—

ENGINE_NAME	PILIER · PHASE	STATUT	PATH	SHA / SIZE	DESCRIPTION FONCTIONNELLE	COUCHES Ω ÉMISES	APIs EXPOSÉES	RISQUES
orchestrator	LEGACY ?	ACTIF_LEGACY	engines/hunt_orchestrator/orchestrator.py	2f78a3f0129eaa56... 17823o	BCE-4X PO — ORCHESTRATEUR DE CHASSE v1	—	—	—
vent_odeurs	LEGACY ?	ACTIF_LEGACY	engines/hunt_orchestrator/vent_odeurs.py	99c387b05625177f... 11395o	BCE-4X PO — ENGINE VENT & ODEURS v1	—	—	—
x6000_product_score	LEGACY ?	ACTIF_LEGACY	engines/nutrition_intelligence/x6000_product_score.py	07bfea2885850076... 11015o	x6000 — PRODUCT_SCORE_ENGINE	—	—	—
candidate_generator	LEGACY ?	ACTIF_LEGACY	engines/relocation/candidate_generator.py	d49a6f3ae5f9ad54... 5885o	BCE-4X BLOC 3 — CANDIDATE GENERATOR	—	—	—
relocation_engine	LEGACY ?	ACTIF_LEGACY	engines/relocation/relocation_engine.py	fb04a55c18191a89... 10885o	BCE-4X BLOC 3 — RELOCATION ENGINE	affuts	—	—
territory_bridge	LEGACY ?	ACTIF_LEGACY	engines/supra_advanced/territory_bridge.py	01638845bee9333b... 5471o	SUPRA-REACT-Omega: Territory IA Data Provider for SUPRA v2	hotspots	—	—
terrain_costs	LEGACY ?	ACTIF_LEGACY	engines/terrain_nav/terrain_costs.py	992ce471479217ba... 11719o	BCE-4X Phase 2.5 — TERRAIN NAV ENGINE (TNE)	—	—	—
terrain_graph	LEGACY ?	ACTIF_LEGACY	engines/terrain_nav/terrain_graph.py	039f120f296a4661... 11787o	BCE-4X Phase 2.5 — TERRAIN NAV ENGINE (TNE)	—	—	—
terrain_router	LEGACY ?	ACTIF_LEGACY	engines/terrain_nav/terrain_router.py	7b7e1ea91844b06d... 20568o	BCE-4X Phase 2.5 — TERRAIN NAV ENGINE (TNE)	—	—	—
terrain_sources	LEGACY ?	ACTIF_LEGACY	engines/terrain_nav/terrain_sources.py	6f1e4d22bfa19c5f... 17054o	BCE-4X Phase 2.5 — TERRAIN NAV ENGINE (TNE)	—	—	—
router	LEGACY V3	ACTIF_LEGACY	engines/weather_v3/router.py	636db94adbdd9cbd... 13020o	Weather Engine v3 — BCE-4X / STEEVE-MAX	—	GET /windgrid GET /hunting-score GET /current	—
wind_model_provider	LEGACY V3	ACTIF_LEGACY	engines/weather_v3/wind_model_provider.py	2ff795863f7ef971... 6889o	BCE-4X — Wind Model Provider Abstraction Layer	—	—	—
__init__	LEGACY ?	INACTIF_LEGACY	engines/bdre/__init__.py	b2412b738ac1b8ef... 3431o	BDRE — BIONIC Data Reliability Engine	—	—	LEGACY non-conforme à BCE-4X
audit_logger	LEGACY ?	INACTIF_LEGACY	engines/bdre/audit_logger.py	0cf0aeae2481bfd... 3292o	BDRE — Audit Logger (F6)	—	—	LEGACY non-conforme à BCE-4X
fallback_chain	LEGACY ?	INACTIF_LEGACY	engines/bdre/fallback_chain.py	21ef92978293ee38... 22722o	BDRE — Fallback Chain (F5)	—	—	LEGACY non-conforme à BCE-4X
quality_scorer	LEGACY ?	INACTIF_LEGACY	engines/bdre/quality_scorer.py	20d8f43df0751516... 7294o	BDRE — Quality Scorer (F2)	—	—	LEGACY non-conforme à BCE-4X
router	LEGACY ?	INACTIF_LEGACY	engines/bdre/router.py	c3daf59360e065ab... 19282o	BDRE — Router (8 endpoints)	corridors, affuts	GET /fallbacks/recent GET /quality/report GET /anomalies/recent	LEGACY non-conforme à BCE-4X
source_registry	LEGACY ?	INACTIF_LEGACY	engines/bdre/source_registry.py	8179d2621f63b089... 8120o	BDRE — Source Registry (F1/F2)	—	—	LEGACY non-conforme à BCE-4X
source_selector	LEGACY ?	INACTIF_LEGACY	engines/bdre/source_selector.py	9b2cc4a065fb9693... 2399o	BDRE — Source Selector (F4)	—	—	LEGACY non-conforme à BCE-4X

	ENGINE_NAME	PILIER · PHASE	STATUT	PATH	SHA / SIZE	DESCRIPTION FONCTIONNELLE	COUCHES Ω ÉMISES	APIs EXPOSÉES	RISQUES
	waterway_classifier	LEGACY ?	INACTIF_LEGACY	engines/bdre/waterway_classifier.py	e6c7ded32715ca4d... 4520o	BDRE — Waterway Classifier (DS-8 Resolution)	—	—	LEGACY non-conforme à BCE-4X
	__init__	LEGACY ?	INACTIF_LEGACY	engines/corridor_unified/__init__.py	0b90008baa28d92c... 111o		—	—	LEGACY non-conforme à BCE-4X
	router	LEGACY ?	INACTIF_LEGACY	engines/corridor_unified/router.py	a91ba0bbfab7c6b5... 7024o	BCE-4X BLOC 1 — CORRIDOR_UNIFIED API	corridors, contamination	POST /build GET /status POST /audit-exclusions	LEGACY non-conforme à BCE-4X
	__init__	LEGACY ?	INACTIF_LEGACY	engines/hunt_orchestrator/__init__.py	3374506dd2048070... 456o	BCE-4X PO — HUNT ORCHESTRATOR ENGINE	—	—	LEGACY non-conforme à BCE-4X
	router	LEGACY ?	INACTIF_LEGACY	engines/hunt_orchestrator/router.py	fc3d4bd8f45f5e41... 16253o	BCE-4X PO — HUNT ORCHESTRATOR API	zones	GET /terrain-cache/stats GET /status POST /scent-zone	LEGACY non-conforme à BCE-4X
	__init__	LEGACY ?	INACTIF_LEGACY	engines/nutrition_intelligence/__init__.py	27d3d1e9d6942326... 1361o	Nutrition Intelligence Engine Package — x5000 SUPRA + x6000 + x6010-x6012	—	—	LEGACY non-conforme à BCE-4X
	ENGINE-V7	LEGACY ?	INACTIF_LEGACY	engines/nutrition_intelligence/router.py	f75762f4e9344fc9... 35742o	API Router — Nutrition Intelligence x5000 SUPRA + x6000 + x6010-x6012	corridors, hotspots	POST /products/compliance POST /products/ecosystem GET /evidence	LEGACY non-conforme à BCE-4X
	x5100_mineral_score	LEGACY ?	INACTIF_LEGACY	engines/nutrition_intelligence/x5100_mineral_score.py	612120c19bf068b0... 5904o	x5100 — MINERAL_SCORE_ENGINE	—	—	LEGACY non-conforme à BCE-4X
	x5200_mineral_recommendation	LEGACY ?	INACTIF_LEGACY	engines/nutrition_intelligence/x5200_mineral_recommendation.py	61acbd99f2fff019... 4286o	x5200 — MINERAL_RECOMMENDATION_ENGINE	—	—	LEGACY non-conforme à BCE-4X
	x5300_order_engine	LEGACY ?	INACTIF_LEGACY	engines/nutrition_intelligence/x5300_order_engine.py	a3342f2e149abab3... 2175o	x5300 — ORDER_ENGINE	—	—	LEGACY non-conforme à BCE-4X
	x5500_energy_protein	LEGACY ?	INACTIF_LEGACY	engines/nutrition_intelligence/x5500_energy_protein.py	a1aa53f156bdaf9b... 15685o	x5500 — ENERGY_PROTEIN_ENGINE	—	—	LEGACY non-conforme à BCE-4X
	x5600_site_guide	LEGACY ?	INACTIF_LEGACY	engines/nutrition_intelligence/x5600_site_guide.py	85e93f9bcd60568e... 9037o	x5600 — SITE_GUIDE_ENGINE	—	—	LEGACY non-conforme à BCE-4X
	x5700_cost_engine	LEGACY ?	INACTIF_LEGACY	engines/nutrition_intelligence/x5700_cost_engine.py	62e1dc0008af9719... 3287o	x5700 — COST_ENGINE	—	—	LEGACY non-conforme à BCE-4X
	x5800_recipe_engine	LEGACY ?	INACTIF_LEGACY	engines/nutrition_intelligence/x5800_recipe_engine.py	1d1593ad02541aec... 3496o	x5800 — RECIPE_ENGINE	—	—	LEGACY non-

ENGINE_NAME	PILIER · PHASE	STATUT	PATH	SHA / SIZE	DESCRIPTION FONCTIONNELLE	COUCHES Ω ÉMISES	APIs EXPOSÉES	RISQUES
								conforme à BCE-4X
x5900_evidence_engine	LEGACY ?	INACTIF_LEGACY	engines/nutrition_intelligence/x5900_evidence_engine.py	7759654fb7669203... 10974o	x5900 — EVIDENCE_ENGINE V2	—	—	LEGACY non-conforme à BCE-4X
x6010_product_quality_analyzer	LEGACY ?	INACTIF_LEGACY	engines/nutrition_intelligence/x6010_product_quality_analyzer.py	7ad548cf41c319d9... 7727o	x6010 — PRODUCT_QUALITY_ANALYZER	—	—	LEGACY non-conforme à BCE-4X
x6011_market_availability_engine	LEGACY ?	INACTIF_LEGACY	engines/nutrition_intelligence/x6011_market_availability_engine.py	544433ede26ba8a0... 12560o	x6011 — MARKET_AVAILABILITY_ENGINE	—	—	LEGACY non-conforme à BCE-4X
x6012_regulatory_compliance_engine	LEGACY ?	INACTIF_LEGACY	engines/nutrition_intelligence/x6012_regulatory_compliance_engine.py	e07d6a282633551e... 11499o	x6012 — REGULATORY_COMPLIANCE_ENGINE	—	—	LEGACY non-conforme à BCE-4X
x6020_terrain_solutions	LEGACY ?	INACTIF_LEGACY	engines/nutrition_intelligence/x6020_terrain_solutions.py	966c41a5c0b3cdd9... 14668o	x6020 — TERRAIN_SOLUTIONS_ENGINE	salines	—	LEGACY non-conforme à BCE-4X
x6030_product_ecosystem	LEGACY ?	INACTIF_LEGACY	engines/nutrition_intelligence/x6030_product_ecosystem.py	3e633d24b3a265be... 12177o	x6030 — PRODUCT_ECOSYSTEM_CONNECTOR	—	—	LEGACY non-conforme à BCE-4X
x7000_supplier_product_engine	LEGACY ?	INACTIF_LEGACY	engines/nutrition_intelligence/x7000_supplier_product_engine.py	515002deb9cfaf2b7... 9953o	x7000 — SUPPLIER_PRODUCT_ENGINE (MongoDB)	—	—	LEGACY non-conforme à BCE-4X
__init__	LEGACY ?	INACTIF_LEGACY	engines/post_smoothing/__init__.py	d8bea42c1665da90... 283o	post_smoothing — Module institutionnel HORS registre V30.	—	—	LEGACY non-conforme à BCE-4X
__init__	LEGACY ?	INACTIF_LEGACY	engines/relocation/__init__.py	70a5257b66fa238a... 121o		—	—	LEGACY non-conforme à BCE-4X
router	LEGACY ?	INACTIF_LEGACY	engines/relocation/router.py	3f826f109d86425f... 4688o	BCE-4X BLOC 3 — RELOCALISATION AUTOMATIQUE API	—	GET /status POST /evaluate	LEGACY non-conforme à BCE-4X
ENGINE-V7	LEGACY V7	INACTIF_LEGACY	engines/spatial_engine_v7/__init__.py	5776153084c2e9b0... 53o		—	—	LEGACY non-conforme à BCE-4X
ENGINE-V7	LEGACY V7	INACTIF_LEGACY	engines/spatial_engine_v7/router.py	adcd1b22918466eb... 30360o	SPATIAL-ENGINE-V7 — Moteur geospatial central	corridors, zones, salines, hotspots	GET /scoring GET /vision-scoring GET /zones	LEGACY non-conforme à BCE-4X
__init__	LEGACY ?	INACTIF_LEGACY	engines/supra_advanced/__init__.py	06bbb37510fa098d... 36o		—	—	LEGACY non-conforme à BCE-4X

ENGINE_NAME	PILIER · PHASE	STATUT	PATH	SHA / SIZE	DESCRIPTION FONCTIONNELLE	COUCHES Q ÉMISES	APIs EXPOSÉES	RISQUES
router	LEGACY ?	INACTIF_LEGACY	engines/supra_advanced/router.py	e59bb9491b2c35b9... 11568o	SUPRA Advanced Engines — BCE-4X / STEEVE-MAX	—	GET /recommendations GET /weather-terrain-correlation GET /risk-assessment	LEGACY non-conforme à BCE-4X
ENGINE-V7	LEGACY V7	INACTIF_LEGACY	engines/supra_engine_v7/__init__.py	f22160ec40507249... 52o		—	—	LEGACY non-conforme à BCE-4X
ENGINE-V7	LEGACY V7	INACTIF_LEGACY	engines/supra_engine_v7/router.py	853c38ba273a9840... 13823o	SUPRA-ENGINE-V7 — Moteur decisionnel central	—	POST /fiche GET /status GET /compare	LEGACY non-conforme à BCE-4X
__init__	LEGACY ?	INACTIF_LEGACY	engines/terrain_nav/__init__.py	ed8a4f58a9e3a795... 7041o	BCE-4X Phase 2.5 — TERRAIN NAV ENGINE (TNE)	—	—	LEGACY non-conforme à BCE-4X
__init__	LEGACY V3	INACTIF_LEGACY	engines/weather_v3/__init__.py	07356eb74fd40746... 31o		—	—	LEGACY non-conforme à BCE-4X